

Exploratory Data Analysis Report

Project 2 | DecodeLabs Industrial Training | Batch 2026

Dataset Overview

Metric	Value
Total Records	1,200 orders
Total Columns	14
Date Range	2023 – 2025
Tools Used	Microsoft Excel, Power BI

1. Problem Statement

The goal of this analysis is to explore the sales dataset to understand product performance across years, identify key referral sources driving customer acquisition, and uncover trends in order status and payment behaviour to support data-driven business decision making.

2. Methodology

The dataset containing 1,200 orders was cleaned in Microsoft Excel, where descriptive statistics and outlier analysis were performed using the IQR (Interquartile Range) method. Power BI was then used to visualize trends and patterns across products, payment methods, referral sources, and order status.

Descriptive Statistics Summary:

Metric	Unit Price	Total Price	Quantity
Mean	356.41	1,053.97	2.95
Median	364.21	823.62	3.00
Min	11.39	11.39	1
Max	699.93	3,456.40	5
Outliers (IQR)	0	0	0

3. Key Findings

Sales Trend: Sales remained stable between 2023 and 2024 at approximately 60K–100K, followed by a sharp decline in 2025. This may be due to incomplete 2025 data representing only a partial

year.

Top Performing Products: Chair and Printer recorded the highest total sales, consistently outperforming other product categories. Laptop and Tablet also showed strong performance.

Lowest Performing Product: Phone recorded the lowest total sales across all years, significantly underperforming compared to all other products in the portfolio.

Order Status Distribution: Out of 1,200 total orders — Cancelled: 250 (20.8%), Returned: 247 (20.6%), Pending: 237 (19.8%), Shipped: 235 (19.6%), Delivered: 231 (19.3%). The 20.8% cancellation rate is a significant business concern.

Payment Methods: All payment methods — Credit Card, Online, Cash, Gift Card, and Debit Card — were used almost equally, indicating diverse customer payment preferences with no single dominant method.

Referral Source: Instagram was the highest performing referral source with ~250 orders, followed by Email and Google. This indicates strong social media influence on purchasing behaviour.

Outlier Analysis: No outliers were detected across Unit Price, Total Price, or Quantity using the IQR method, confirming the dataset is clean and reliable for further analysis.

4. Power BI Dashboard

The following dashboard was built in Power BI to visually explore trends and patterns across all key dimensions of the dataset.

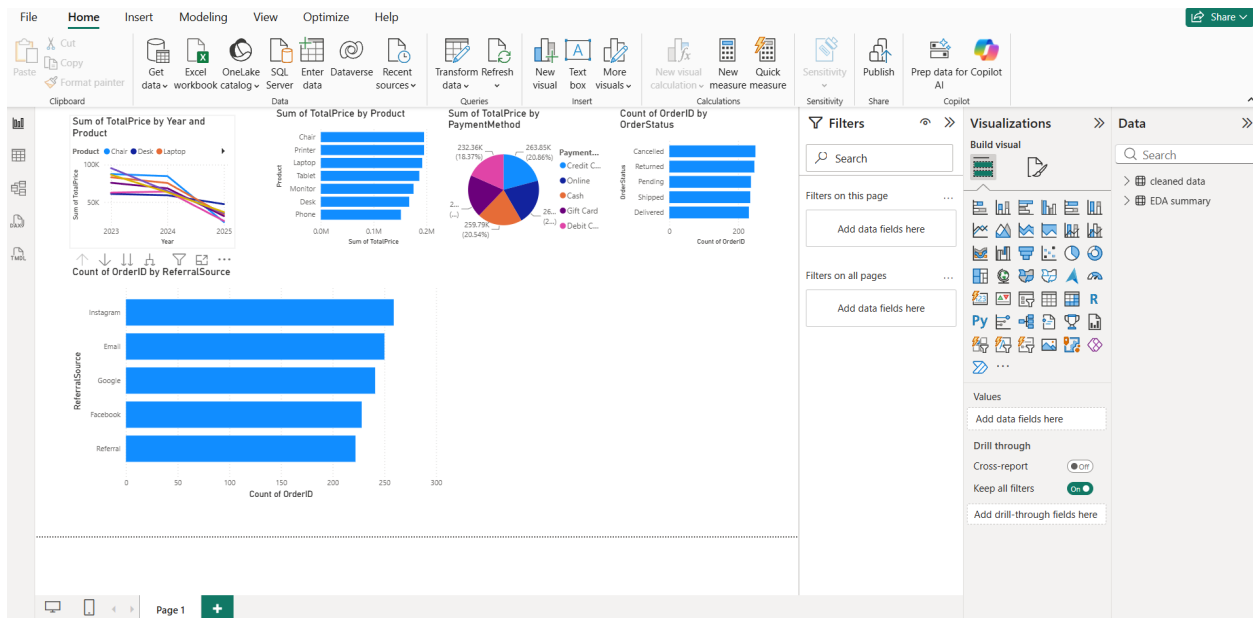


Figure 1: Power BI Dashboard — Sales Trends, Product Performance, Payment Methods, Order Status & Referral Sources

5. Recommendations

1. Investigate why Phone sales are underperforming and introduce promotional offers or targeted discounts to boost its performance.
2. Analyze the reasons behind the 20.8% cancellation rate and implement better customer communication, improved logistics, and clearer return policies.
3. Increase the marketing budget on Instagram as it is the highest performing referral source to drive more customer orders.
4. Investigate the sharp sales decline in 2025 by examining seasonal patterns and external market factors before drawing final conclusions.
5. Maintain support for all payment methods since customers show equal preference — removing any option could negatively impact conversions.